

Detailed Test Results - Milestone 1

Project: Rolling Loans: Continuous Borrowing on Danogo

Milestone 1: Smart Contract Development

Evidence scope: detailed test results, technical report describing test scenarios, testnet deployment transaction evidence, and Preview/Preprod transaction IDs demonstrating functionality.

1. Test Result Summary

The table below presents the detailed test results mapped to actual transaction evidence. The wording intentionally distinguishes manual/external repayment from the atomic settlement that occurs inside the refinancing transaction.

Note on Aiken test coverage

The Aiken file tests/rolling_loan_test.ak contains fixture-based unit tests for the rolling-loan mechanism. These tests validate the core constraints behind the Milestone 1 transaction evidence, including Create Loan + Repay Loan refinancing, interest-index update, collateral continuity, pool accounting, token consistency, utilization cap, and invalid-state rejection.

The values in the Aiken tests are fixture values used for unit testing. The Preprod transaction evidence in this report demonstrates the same rolling/refinancing flow with actual on-chain values. Therefore, the PDF evidence and the Aiken test file are complementary: the PDF summarizes the public on-chain transaction proof, while the Aiken file documents the fixture-based unit test coverage for the same mechanism.

Aiken test / group	Supports evidence items	Purpose
succ_rolling_loan_full_repay_and_create_new_loan	T01, T02, T03, T08	Validates the successful Create Loan + Repay Loan refinancing flow in a single transaction.
succ_interest_recalculation_after_time_elapsed	T06	Validates interest-index update after elapsed time.
succ_rolling_loan_interest_index_correctly_reset	T06	Validates that the new loan starts from a refreshed interest baseline.
succ_rolling_loan_with_multiple_collaterals_transferred	T07	Validates collateral continuity / transfer across the refinancing flow.
succ_rolling_loan_with_zero_origination_fee	Refinancing edge case	Validates the refinancing flow when origination fee is zero.
Failure / constraint tests	Constraint coverage	Validate rejection cases for interest index, collateral sufficiency, pool accounting, token mismatch, utilization cap, minimum transaction amount, and pool reserve constraints.

Test ID	Scenario	Expected result	Actual evidence	Status
T01	Source active loan exists in source loan contract	An existing active loan UTxO is present and available to be consumed by the rolling-loan transaction.	Input UTxO fc25ba0804c0fc8cb2552b97695b6ea6ec64f21551bebccd49e71b5c2abb2249#1 is consumed by tx b09fa33a42fda235b28be38b58f1daab764e3dfb58768b9506a3c9d5ba5e851f.	PASS
T02	Create target loan position	A new loan position is created in the target loan contract within the same transaction.	Output UTxO at address addr_test1zpk4s8x6yjl8wspfqheqdgtr89dzemyrpqyc7n5my9v8027kk6djsz5vkduazxj0d7avth6k52cqy0ld77y64tpnlyejsp2fur is created by tx b09fa33a42fda235b28be38b58f1daab764e3dfb58768b9506a3c9d5ba5e851f.	PASS
T03	Cross-contract refinancing / migration	The transaction moves the loan state from Float Loan SC to Leverage Loan SC through one atomic transaction.	Cardanoscan transaction message shows Migrate Loan and Create Loan; UTxO input/output analysis shows consumed source loan and created target loan.	PASS
T04	Borrowed token continuity	The source and target loan states should reference the same borrowed token.	Before token: fUSDm. After token: fUSDm.	PASS
T05	Loan amount carried forward and updated	The target loan amount should reflect the new loan state after refinancing/settlement.	Before loan amount: 50,000,000. After loan amount: 50,011,310.	PASS
T06	Interest baseline reset / updated	The target loan should start with an updated interest baseline to avoid double-counting old interest.	Before initial interest index: 1,010,292,917,893. After initial interest index: 1,061,924,584,989.	PASS
T07	Collateral continuity	Collateral should not be withdrawn to the user wallet before opening the new position; it should be redirected to the new position.	Both before and after loan UTxOs show fETH 100,000,000 as part of the loan-position asset set.	PASS
T08	No separate manual repayment before rollover	The borrower should not need to manually close the old loan in a separate transaction before creating the new loan.	The single rolling-loan transaction performs Create Loan + Repay Old Loan + Collateral Transfer atomically.	PASS

2. Before/After Datum Comparison

Field	Before migration: consumed source loan UTxO	After migration: created target loan UTxO	Result
Contract role	Source Loan UTxO consumed by migration transaction	Target Loan UTxO created by migration transaction	PASS

Field	Before migration: consumed source loan UTxO	After migration: created target loan UTxO	Result
Network	Cardano Preprod testnet	Cardano Preprod testnet	PASS
Address	addr_test1zzx7xnchq5lhgj63z3yl0f8xk3v9vshd3qd6kgmnsn3kavfxk6djsz5vkduazxj0d7avth6k52cqy0ld77y64tpnlyejss8ekch	addr_test1zpk4s8x6yjl8wspfqheqdgtr89dzemyrpqyc7n5my9v8027kk6djsz5vkduazxj0d7avth6k52cqy0ld77y64tpnlyejsp2fur	PASS
Datum hash	f25a595b47d11117506bcd03cbc6af289a0eed565d8a53b04f9e3e1f02de3db	6025de088b25d7175d75f4747adf269287028e1fa616dde709b0598fa1a13c0e	PASS
Loan identifier asset	8de34f17053f744b511449f7a4e6b4585642ed881bab23709c6dd624 / f1fedb6d6ae956bfa00ec958279720d299d5d2b470fd1595229fd95e	6d581cda24be77402905f206a163395a2cec8308098f4e9b290ef57a / a74fe9b60303cc4633e7037cd1b376c22ca03d7abc1be0f503333cb5	PASS
Borrowed token	834a15101873b4e1ddfaa830df46792913995d8738dcde34eda27905 / fUSDM	834a15101873b4e1ddfaa830df46792913995d8738dcde34eda27905 / fUSDM	PASS
Loan amount	50,000,000	50,011,310	PASS
Initial interest index	1,010,292,917,893	1,061,924,584,989	PASS
Collateral shown in UTxO	fETH 100,000,000 plus loan/position assets	fETH 100,000,000 plus loan/position assets	PASS

Conclusion

The before/after datum comparison shows that the transaction consumes a source loan state and creates a target loan state. The borrowed token remains fUSDM, the loan amount is updated, and the interest baseline is reset/updated. This demonstrates the core rolling-loan/refinancing mechanism at the smart contract state-transition level.